

Brief Contents

- 1** Evolution, the Themes of Biology, and Scientific Inquiry 2

Unit 1 THE CHEMISTRY OF LIFE 27

- 2** The Chemical Context of Life 28
3 Water and Life 44
4 Carbon and the Molecular Diversity of Life 56
5 The Structure and Function of Large Biological Molecules 66

Unit 2 THE CELL 92

- 6** A Tour of the Cell 93
7 Membrane Structure and Function 126
8 An Introduction to Metabolism 143
9 Cellular Respiration and Fermentation 164
10 Photosynthesis 187
11 Cell Communication 212
12 The Cell Cycle 234

Unit 3 GENETICS 253

- 13** Meiosis and Sexual Life Cycles 254
14 Mendel and the Gene Idea 269
15 The Chromosomal Basis of Inheritance 294
16 The Molecular Basis of Inheritance 314
17 Gene Expression: From Gene to Protein 335
18 Regulation of Gene Expression 365
19 Viruses 398
20 DNA Tools and Biotechnology 415
21 Genomes and Their Evolution 442

Unit 4 MECHANISMS OF EVOLUTION 467

- 22** Descent with Modification: A Darwinian View of Life 468
23 The Evolution of Populations 486
24 The Origin of Species 506
25 The History of Life on Earth 525

Unit 5 THE EVOLUTIONARY HISTORY OF BIOLOGICAL DIVERSITY 552

- 26** Phylogeny and the Tree of Life 553
27 Bacteria and Archaea 573
28 Protists 593
29 Plant Diversity I: How Plants Colonized Land 618
30 Plant Diversity II: The Evolution of Seed Plants 636
31 Fungi 654
32 An Overview of Animal Diversity 673
33 An Introduction to Invertebrates 686
34 The Origin and Evolution of Vertebrates 718

Unit 6 PLANT FORM AND FUNCTION 757

- 35** Vascular Plant Structure, Growth, and Development 758
36 Resource Acquisition and Transport in Vascular Plants 784
37 Soil and Plant Nutrition 805
38 Angiosperm Reproduction and Biotechnology 822
39 Plant Responses to Internal and External Signals 842

Unit 7 ANIMAL FORM AND FUNCTION 872

- 40** Basic Principles of Animal Form and Function 873
41 Animal Nutrition 898
42 Circulation and Gas Exchange 921
43 The Immune System 952
44 Osmoregulation and Excretion 977
45 Hormones and the Endocrine System 999
46 Animal Reproduction 1019
47 Animal Development 1043
48 Neurons, Synapses, and Signaling 1067
49 Nervous Systems 1085
50 Sensory and Motor Mechanisms 1107
51 Animal Behavior 1139

Unit 8 ECOLOGY 1163

- 52** An Introduction to Ecology and the Biosphere 1164
53 Population Ecology 1190
54 Community Ecology 1214
55 Ecosystems and Restoration Ecology 1238
56 Conservation Biology and Global Change 1260